

Burmese-English Machine Translation Model Report
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Introduction

Translating Burmese to English poses major structural challenges because Burmese is a morphologically rich, Subject-Object-Verb (SOV) language without word spaces, unlike English's Subject-Verb-Object (SVO) structure (Win & Nwe, n.d.). Consequently, standard machine translation systems suffer from severe data sparseness and out-of-vocabulary (OOV) rates exceeding 50% when trained on limited parallel corpora, resulting in distorted translations and communication barriers (Zin et al., n.d.). Empirical research demonstrates that addressing morphological complexity and structural differences significantly mitigates OOV errors, raising overall translation F-scores from 72.34% to 80.2% (Zin et al., n.d.). To address this, we will fine-tune a pre-trained sequence-to-sequence Transformer model on a specialized Burmese-English parallel corpus, using self-attention and custom tokenization to transform Burmese SOV inputs into accurate, fluent SVO English outputs (*Machine Translation in AI*, 2026).

Dataset

To fine-tune the model, we selected the open-source Myanmar-English General Text Translation dataset. Developed by Khant Sint Heinn (Kalix Louis), an NLP and machine learning engineer and the Lead Developer at DatarrX, it aims to advance open-source AI development for low-resource languages (kalixlouiis, 2026). It is distributed under the Creative Commons Attribution-NonCommercial-ShareAlike 4.0 International License (CC-BY-NC-SA 4.0).

This dataset was chosen due to its manageable size: 12,021 parallel sentence pairs, with 9,616 samples (~9.6k or 80%) allocated for training, 10% for validation, and 10% for testing (Kalixlouiis, 2026). Furthermore, it encompasses a wide-ranging general domain, including everyday situations, literary works, and conversational narratives (Kalixlouiis, 2026). Additionally, the corpus is of high accuracy, robustness, and exceptional quality, having undergone thorough duplicate removal, structural cleaning, and human verification to ensure precise semantic alignment and a natural, engaging tone (Kalixlouiis, 2026).

Note: Due to time and resource limits, only a small portion of the dataset (100 training samples, 10 validation samples, and 10 testing samples) is used.

Fine-tuning NLLB-200-1.3B-8bit

Fine-tuning the NLLB-200-1.3B-8bit ([Emilio407/nllb-200-1.3B-8bit](#)) sequence-to-sequence model on a single Google Colab T4 GPU (15GB VRAM) requires combining Quantization, Parameter-Efficient Fine-Tuning (PEFT), and optimized batching (Godoy, 2025).

1. Preventing CUDA Out-of-Memory (OOM) via Quantization

When fine-tuning a 1.3-billion-parameter model in full 32-bit floating-point (FP32) or 16-bit (FP16) precision without quantization, the GPU must store not only the model weights (~2.6 GB–5.2 GB) but also the gradients, optimizer states (such as AdamW), and intermediate layer activations (Godoy, 2025). On a 15GB T4 GPU, these requirements quickly exceed available VRAM during the backward pass, triggering a CUDA out-of-memory (OOM) error (Godoy, 2025).

By loading the base model in 8-bit or 4-bit quantized precision via `BitsAndBytesConfig`, the memory footprint of the core model weights is drastically reduced (*The Complete Guide to LLM Quantization*, 2025). The base quantized weights are then frozen, ensuring that full-precision gradients and optimizer states are not computed for all 1.3 billion parameters (Godoy, 2025).

2. Parameter-Efficient Fine-Tuning (PEFT) and Rank Selection ($r=8$)

To update the model without modifying the frozen base weights, Low-Rank Adaptation (LoRA) attaches small, trainable adapter matrices to targeted layers (`k_proj`, `q_proj`, `v_proj`, `out_proj`, `fc1`, `fc2`) (Godoy, 2025):

- **Why Rank 8 ($r=8$)?** The rank (r) determines the inner dimension of the low-rank update matrices (Godoy, 2025). Selecting $r=8$ provides an optimal balance between representation capacity and memory efficiency. It keeps the number of trainable parameters extremely small (typically ($<1\%$) of total model parameters), preventing memory overflow on the T4 GPU while providing sufficient expressive power to adapt NLLB-200 to the Burmese-to-English translation task.
- **LoRA Scaling (`lora_alpha=16`):** Setting `lora_alpha=16` serves as a scaling factor (typically $2r$), stabilizing learning updates relative to the adapter rank (Godoy, 2025).
- **K-Bit Preparation:** Before attaching adapters, calling `prepare_model_for_kbit_training(model)` ensures numerical stability during quantized training (Godoy, 2025).

3. Batch Training for Training Efficiency

Processing sentences individually introduces high computational overhead and underutilizes GPU parallel hardware (Clark, n.d.). Batch training optimizes performance on the T4 GPU through two key techniques:

- **Batch Size Selection:** Setting a per-device batch size (e.g., `per_device_train_batch_size=8`) fits comfortable micro-batches into VRAM alongside adapter gradients (Godoy, 2025).

- **Dynamic Padding** (`DataCollatorForSeq2Seq`): Rather than padding every sequence in the dataset to a fixed global maximum length (e.g., 128 tokens), `DataCollatorForSeq2Seq` dynamically pads sentences in each mini-batch to match only the longest sentence in that specific batch (*Causal Language Modeling*, n.d.). This eliminates unnecessary padding tokens, maximizing matrix multiplication throughput and accelerating per-epoch training speeds (*Causal Language Modeling*, n.d.).

Evaluation

SacreBLEU is a standardized implementation of the BLEU (Bilingual Evaluation Understudy) metric to evaluate machine translation by measuring n-gram overlap between model-generated translations and human reference translations (*BLEU and ROGUE score for NLP evaluation*, 2026). Higher SacreBLEU scores generally indicate greater similarity to the reference translations (*BLEU and ROGUE score for NLP evaluation*, 2026). During fine-tuning, SacreBLEU is calculated on the validation set after each epoch to monitor changes in translation quality. The implementation decodes generated predictions and reference labels before calculating the BLEU score.

After fine-tuning, the model is evaluated using the held-out test set, which is separate from the training and validation data. The same test samples and SacreBLEU calculation are used for both models to ensure a consistent comparison.

The original NLLB-200-1.3B-8bit model serves as the baseline, while the LoRA-adapted model represents the fine-tuned system. The baseline model achieved 6.4056, whereas the fine-tuned model scored 6.4576. Their SacreBLEU scores are compared to quantify whether fine-tuning on the Burmese-English parallel corpus improves translation performance. A higher score for the fine-tuned model indicates improved agreement with the reference translations.

Future Plan

Future work will focus on improving translation quality by expanding and diversifying the Burmese-English parallel corpus and conducting more extensive fine-tuning experiments. Different LoRA configurations, learning rates, batch sizes, and training epochs can be evaluated to identify more effective settings. Additional evaluation on unseen and domain-specific Burmese text can also be conducted to assess the model's generalization and practical translation performance.

Summary

This project fine-tunes the quantized NLLB-200-1.3B model for Burmese-to-English machine translation using PEFT with LoRA, enabling training within the limited GPU memory available on Google Colab. The Myanmar-English General Text Translation dataset provides parallel sentence pairs for training, validation, and testing. SacreBLEU evaluation enables an

objective comparison between the base and fine-tuned models to determine the effectiveness of the adaptation process.

Reference

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